# MS-ML Studio: A Generic Free and Open-Source Tool for Mass Spectrometry Analysis and Classification

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**Abstract**

Mass spectrometry is a useful tool for biochemistry research. It has been used to identify microorganisms, proteins, chemical compounds and as a diagnosis of many diseases by blood proteins, lipids and carbohydrate composition analysis. Several tools and libraries are available to handle, interpret and classify mass spectrometry (MS) data. However, user friendly computational tools are lacking the capability of integrating all the steps in mass spectrometry data analysis. Here we report the development of a free machine learning solution capable of handling all steps in MS data analysis from raw spectra mass interpretation to classification.

Results: We developed a free and open source MS-ML (Mass Spectrometry Machine Learning ) Studio and tested its ability on feature extraction of different mass spectrometry data and classifying with conventional MLP algorithm, embedded on MS-ML; we also export those features to Weka to improve the performance of classification task with other algorithms. The accuracy of MS-ML Studio using internal MLP algorithm on classifying features extracted from data derived from Aeromonas intact bacterial cells reaching accuracies in the range of 89% to 99%.

*Keywords:*
Machine Learning, Feature Extraction, Mass Spectrometry, Peak Selection, Biomarker Finder, MS-ML Studio, Feature extraction

## 1. Introduction

Mass Spectrometry (MS) analysis has widespread applications in biochemistry research field. This technique allows the accurate determination of molecular mass of chemical species; such information can be used to identify chemical compounds in complex mixtures. MS can also be applied to diagnose diseases such as cancer and to identify pathogenic bacteria.

Arnold *et al.* [1] used the LC-MS/MS ion trap®spectra and machine learning to predict peptides fragmentation based on spectra pattern using an automatic computational approach. The rapid clinic identification of pathogenic and food spoilage microorganisms is needed since the actual techniques are relatively slow compared with mass spectrometry approaches[2]; According to Krishnamurthy and Ross (1996) whole-cell analysis has some advantages over chemotaxonomic markers, the proteins found in bacteria are great in abundance and it provide an indirect information about genetic data[3]. Croxato, Prod'Hom and Greub (2012) say that mass spectrometry (MALDI-TOF MS) devices revolutionized the microorganisms identification allowing accurate routine identification at species level of gram-positive and gram-negative bacterial strains; this technique is rapid, high throughput, low cost and efficient identification[4].

Figure 1 Wagner,Naik and Pothen.[5] describes the biomarker discovery process by MS.

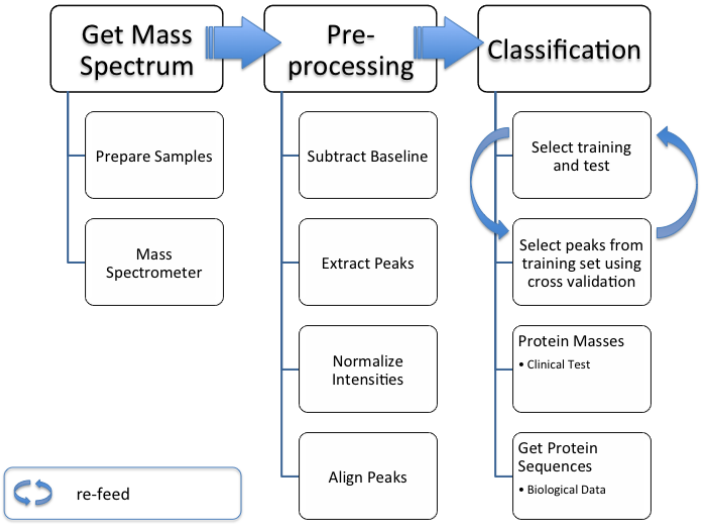



Figure 1: Abstract workflow shows three main steps for biomarker discovery using mass spectrometry adapted form Wagner M. *et al* [5].

For this research we focus in silico analysis, pre-processing and classification steps represented on figure 1, approaching correlation based feature extraction strategies.

There are specific approaches for each cases; mass spectrometer like Ciphergen ProteinChip®(Ciphergen Biosystems, Freemont, CA, USA) have its own proprietary software for preprocessing tasks [6] likewise MALDI Biotyper®(Bruker Daltonics,Bilerica, MA) for microorganisms identification [7].

| Programs | Smooth | Base Line Correction | Peaks Selection |
|---|---|---|---|
| Cromwell | Undecimated Discrete Wavelet Transform | Monotone minimum | SRN,Local maximum |
| LCMS-2D | | Moving average of minima | SRN,Detection/Intensity threshold |
| LIMPIC | Kaiser window | Linear interpolation | SRN,Slopes of peaks |
| LMS | Gaussian filter | Linear interpolation | SRN,Local maximum |
| MapQuant | Moving average filter,Savitzky-Golay filter,Gaussian filter | | Model-based criterion |
| CWT | Continuous Wavelet Transform | Continuous Wavelet Transform | SRN,Ridge lines |
| msInspect | Discrete Wavelet Transform | Linear interpolation,Loess | Shape ratio |
| mzMine | Moving average filter,Savitzky-Golay filter | Monotone minimum | SRN,Detection/Intensity threshold,Peak width |
| OpenMS | Continuous Wavelet Transform | | Model-based criterion |
| PROcess | Moving average filter | | SRN,Detection/Intensity threshold,Shape ratio |
| PreMS | Undecimated Discrete Wavelet Transform | | SRN,Local maximum |
| XCMS | Gaussian filter | | SRN,Local maximum |

Table 1: Shows a set of software and Smoothing, Base line Correction and peaks selection strategies used to pre-process Mass Spectrometry data, adapted form Yang, He and Yu[8] .

Most of those mass spectrometers provide a way to export the raw data and other formats. Table 1 shows some of the available mass spectrometer independent software and algorithms for preprocessing of MS data. For Wagner,Naik and Pothen. [5], the accuracy of classification depends on how MS data is preprocessed.

According Yang, He and Yu [8] the smooth filter usually apply traditional signal process techniques as moving average and Wavelet transform.Pluskal *et al.*[9] use variation of "Gausian Filter" to remove "shoulder peaks" on high dimensional data. For Yang, He and Yu [8] base line correction methods is typically a two-step process: Estimate the baseline and subtract the baseline from the signal; for example, the strategy used by Wagner,Naik and Pothen [5] generate baseline correction for different intervals of mass spectra, the intervals and baseline parameters is expressed on Table 2 and then subtracts for each window specific base line correction.

| *m/z* (kDa/z) | 1.7–2.3 | 2.3–3.7 | 3.7–10 | 10–15 | 15–40 | 40–60 | 60–100 | 100–205 |
|---|---|---|---|---|---|---|---|---|
| smoothing parameter | 1% | 2% | 5% | 10% | 20% | 40% | 50% | 70% |

Table 2: The baseline correction parameters for window intervals used for preprocessing lung cancer data . (Author:Wagner,Naik and Pothen. [5])

Peaks alignment is an optional step, depending on used mass spectrom-

eter platform; some platforms provide embedded software that make peaks alignment during the process of mass spectrometry. Icoshift, developed for Tomasi *et al.* [10], provided as an open source library used originally for align of chromatographic data, use gap insertion to align more relevant peaks on a selected area. Random Sample Consensus (RANSAC) aligner algorithm used on MZmine2 developed by Plustal *et al.* [9], uses an evolutionary probabilistic approach that on each interaction select a random subset to fit the alignment model; RANSAC algorithm should to be executed after peaks selection process. For this case, RANSAC algorithm uses selected peaks instead processing full mass spectrometry data.

The main problem of peaks selection is the noise presence on spectra. When you have a clean spectra like expressed on figure 2, a simple threshold line may be used as peaks selection task. Many of Table 1 algorithms for peaks selection are derived from cut-line threshold technique.

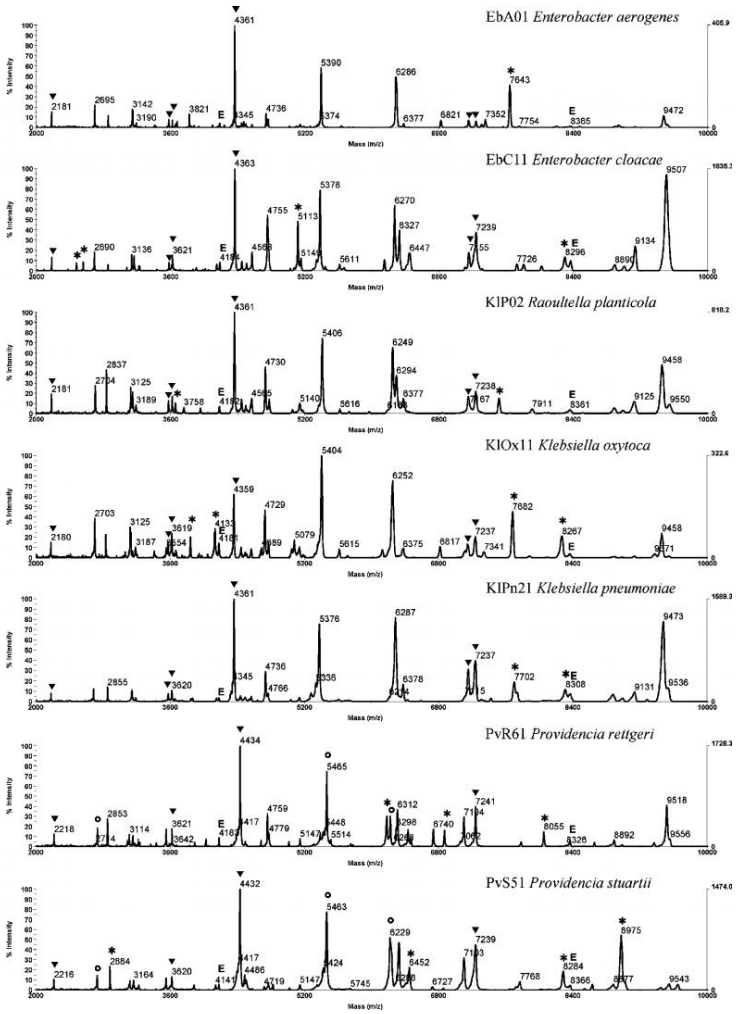



Figure 2: Intact cell strategy showing the selected peaks for different microorganisms ( Author: Bohme et. al. 2010 [2]).

Armananzas et. al.[11] propose Peakbin to handle peaks selection focusing on biomarker finding; this approach use Univariate Marginal Distribution Algorithm (UMDA), an evolutionary algorithm that selects more discriminant peaks for specific sample subset. MZmine 2[9] use multiple methods to select peaks: the "Local Maximum" uses an recursive algorithm that looks for spectrum local maximum; the "wavelet transform", which is suitable for noisy data, and match the peaks mass and charge values to the "Mexican Hat" wavelet model.

Peaks selection step is the purpose of use different techniques on preprocessing phase. MZmine 2 [9] have a set identification strategies, accessing public databases:

- HMDB (Human Metabolome Data Base), that in 2007, according Wishart *et al.*[12], was considered as largest and most comprehensive organisms-specific metabolomics database and it is widely used to characterization and rationalization of biomarkers.
- KEGG [13] (Kyoto Encyclopedia of Genes and Genomes), that consists into three databases: PATHWAY for molecules interaction; GENES for partial and complete genome and LIGAND for collection of chemical compounds in the cell, enzyme and enzymatic reactions.
- METLINE ( Metabolite Database )[14] its objective is provide structural and physical data on know endogenous metabolites, drugs and drugs metabolites; high-accuracy FTMS (Fourier Transform Mass Spectrometry); reference tandem MS data from known metabolites and metabolites derivate and LC/MS profiles from primarily human and some model organisms.
- PubChem [15] is public repository for biological properties of small properties.

Exist other similar free access databases on the web, like DrugBank 4.0 [16] that comprehends more than one thousand and two hundred (1200) biochemical and pharmacological information about drugs, their mechanisms and their targets. There are many other databases that may be used to handle mass spectrometry data analysis. These databases have a large datasets and may not be useful to download; then internet access is always necessary; The classification task is another way to handle mass spectrometry data. For colorectal cancer analysis, use Genetic Algorithm (GA) to biomarker discovery approach and Bayes classifier with K fold cross validation to perform classification task for two classes, cancer and not cancer [17]. Sub-vector machine (SVM) was also used with K-fold cross validation thousand (1000) times to classify ovarian cancer data-set[18].

## 2. Materials and Methods

We develop a generic tool capable to handle different types of Mass Spectrometer; we use as basis provided Aeromonas[19] data set.

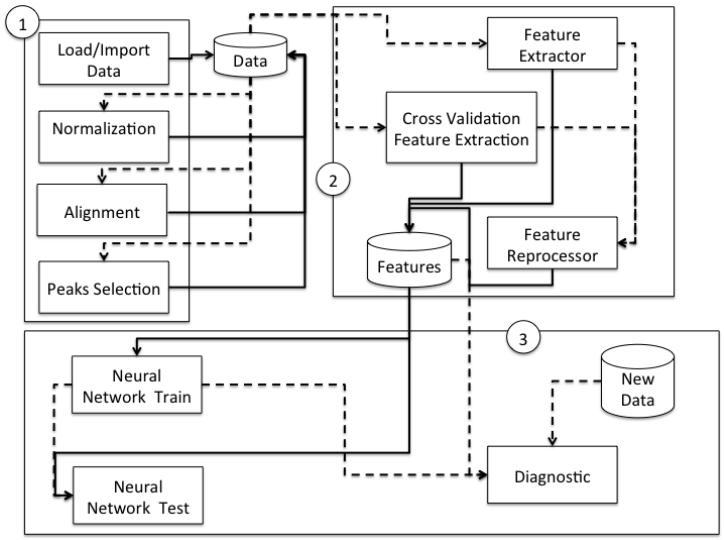



Figure 3: ML-MS workflow is divided in three main general section. The section 1 represents data pre-processing where all data is normalized all peaks aligned and selected. The section 2 represent all types of feature extraction that can be used. The section 3 is about machine learning and the consolidation of diagnostic tool.

The figure 3 represents all steps of ML-MS workflow. The next sections explain each of those steps in details.

### *2.1. Preprocessing*

For ML-MS Studio importation file consists in raw data to shape the system definitions, the figure 4 show an example of raw data where the first column represent the mass/charge values and the second column represent the relative intensity.the Figure 3.1 is formed by four main steps. The import/load data is the beginning of preprocessing data, this step bring the raw data into our system format and start to work with these data, after imported all data can be saved by ML-MS Studio on your own format.

```
2997.935128 1763.000000
2998.270737 1822.000000
2998.606366 1870.000000
2998.942013 1910.000000
2999.277679 1945.000000
2999.613364 1733.000000
2999.949068 1854.000000
3000.284790 1833.000000
3000.620531 1831.000000
3000.956291 1728.000000
3001.292070 1792.000000
3001.627868 1949.000000
3001.963684 1879.000000
3002.299519 1806.000000
3002.635373 1923.000000
3002.971246 1720.000000
3003.307137 1659.000000
3003.643048 1749.000000
3003.978977 1617.000000
3004.314924 1594.000000
3004.650891 1468.000000
3004.986876 1592.000000
```

Figure 4: Raw data file used as input for ML-MS Studio is a text file with two columns separated by a blank character ( space ) where the first column represent the '$M/Z$' values and the second one represent the relative intensity.

There is no function on ML-MS Library to convert conventional data, like mzml or xml, to raw data.

*2.1.1. Normalization and Re-shaping*

High resolution spectra increase the time to process the data by the machine and consequently the complexity of analysis. To solve this problem and satisfy all workflow problems we create 2 buttons on MS-ML Software: the "Normalization" and the "Re-shaping" button; both processes are based on "fn_norm_max_min" on ML-MS Library, this function normalize a provided matrix or a vector values between zero and one.

The normalization button is relative to intensity values of spectra and for each spectra we apply the following equation:

$$N(x, max, min) = (\ x_i - \min\ )\ /\ (\ max\ -\ \min\ ) \tag{1}$$

Where 'x' is the intensity vector and 'i' represents each position on spectra. The re-shaping strategy is about of normalize of $m/z$ values on specific range interval. There is two reasons for normalizing $m/z$ values: Some spectra data have different $m/z$ range and for math purposes operations with different sizes vectors is not possible. The figure 5 show two optional solution for spectra with different sizes using the minimum size or accepting the max size and fill the lacks with zeros, for intensity vectors.

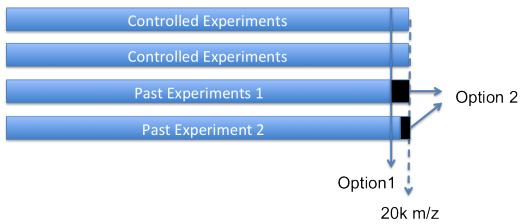



Figure 5: Normalization problems may occur with different spectra sizes, considering blue bars as the $m/z$ range representation; considering the controlled experiment as the current experiment and two other past experiments as older experiments with different sizes that can be used with the same study. The researcher have 2 options use only the smaller $m/z$ range, figure "option 1", or force use the bigger $m/z$ size filling the the figure "black spaces" with zeros.

Other advantage of using re-shaping is the dimensional reduction and alignment need is also reduced. The figure 6 shows problems with amount of data on $m/z$ interval and how reshaping will resolve this problem.

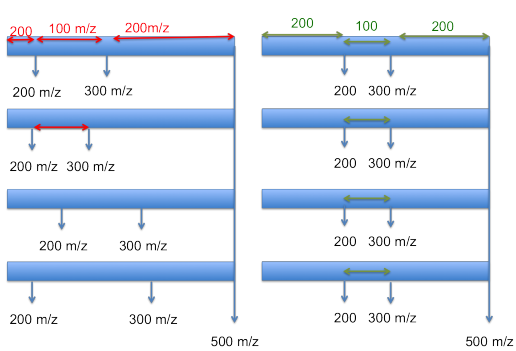



Figure 6: This figure shows how reshaping helps the alignment of information using the $m/z$ value to fit the correct ammount of data to interval with same size.

To reshape the user must use the MS-ML Library function called 'fn_reshape'. This function receive essentially the data and the size of new spectrum 'shape'. Function 'fn_reshape' use $m/z$ values and normalize between max and min $m/z$ values of current experiment. The following equation explains mathematically the reshaping procedure:

$$G(x; W, max, min) = round(W * N(x; max, min)); \tag{2}$$

and:

$$G(x; W, max, min) = (1111112222222233333333...W) \tag{3}$$

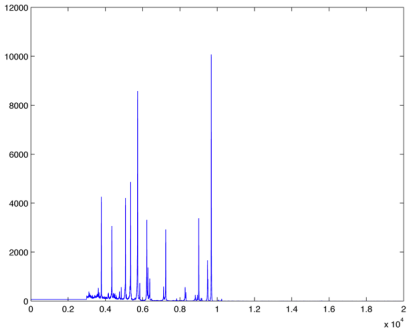
Having 'W' as the new spectra resolution and 'N' the normalization function. For each spectra consider the following algorithm:

Let: $sii$ be a intensity vector;
$smz$ be a $m/z$ vector;
$gmax$ be the maximum $m/z$ value;
$gmin$ and minimum $m/z$ value;
$nmz$ be the new $m/z$ vector;
$nii$ be the new intensity vector;
$W$ be the new spectrum resolution;
Output: $nmz, nii$;

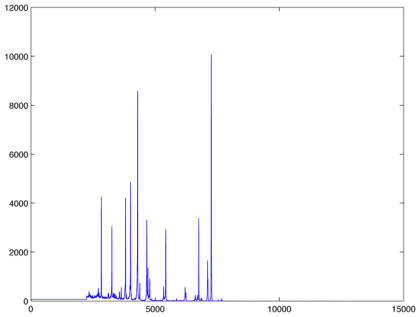
```
g = G(smz, W, gmax, gmin);
gmx = max(g);
for i = 0 to gmx do
    gidx = find(q == i);
    nii(i) = max(sii(gidx));
    nmz(i) = mean(smz(gidx));
end
```

**Algorithm 1:** Representation of reshaping algorithm with output variables $nii$ and $nmz$.

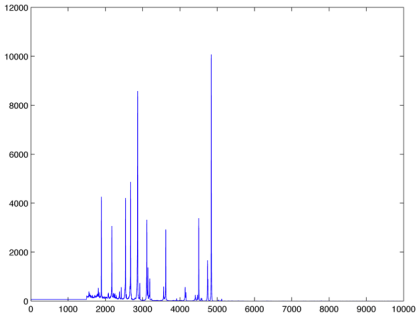
The figure 7 sub-images are the representation of algorithm 1 execution with $W = 1K, 5K, 10k, 15K$ and $20k$.



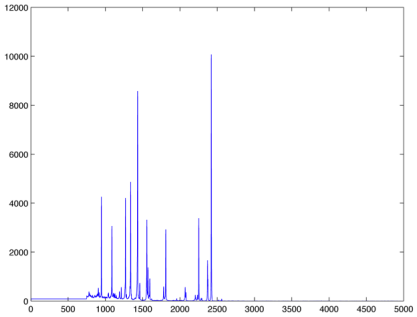

(a) 20k



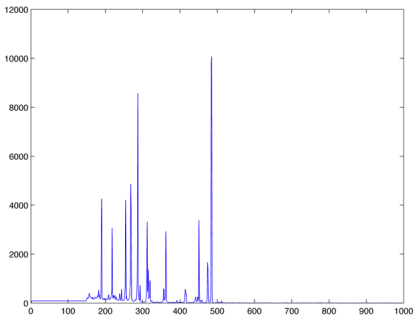

(b) 15k



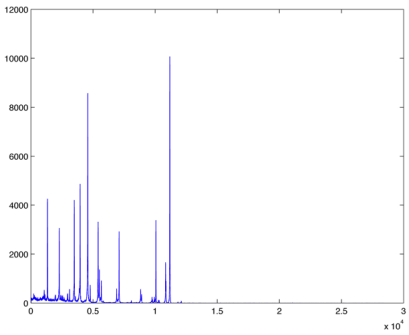

(c) 10k




(d) 5k




(e) 1k




(f) original spectra

Figure 7: 28k values spectra ( f ) reshaping with different resolutions. Only the 1k reshaping (e) is possible to see some few differences form original spectrum format, considering the spectra shape not the peaks positions.

Is important to note that reshaping may impact on peaks selection parameters because of density of data; configuration for higher resolution spectrum does not work in lower resolution because of sensibility of parameters for specific amount of data. The figure 7 shows original spectrum (figure 7.f) reshaped to different sizes. Note that the reshaped spectrum have a lack of peaks that do not happen on original spectrum; the chosen spectrum have one peak with 0 *m/z*; there is no problem if only one spectrum have this information, the reshaping algorithm use the minimum and maximum values of all spectra.

*2.1.2. Peaks Selection*

The peaks selection strategy is the first step for MS analysis; on this step the real peaks are differed from the spectrum noise. The fuzzy moving average (FMA) strategy can be set to be specific, selecting the smaller peaks, or general, selecting only more expressive peaks; different configurations on peaks selection can impact on correlation on automatic feature extraction process.

Considering that 'x' is all intensity values on spectrum the peaks selection process is leaded by following equations:

$$S(x; p, w) = \frac{\sum_{i=p-w}^{p+w}(x_i T(2w+1, i))}{2w+1} \tag{4}$$

having:

$$p \epsilon N^* |\ 1 \leq p - w,\ p + w \leq S \tag{5}$$

Where 'p' is the *m/z* vector index; 'w' is the distance from the window edges to the center represented on figure 8 ; and 'x' is the vector of intensities; 'i' represents the window index and 'T' is an triangular function that receive the size and the respective window indexer 'i'.

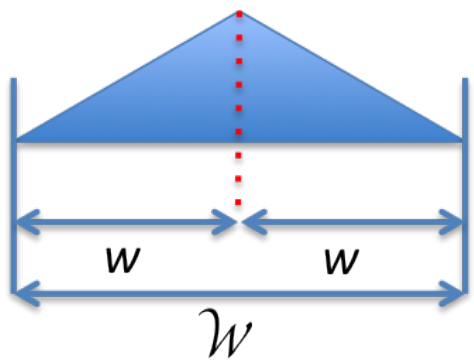



Figure 8: Window representation of FMA.

The figure 9 is the representation of FMA with windows sizes of 10, 100, 200 and 500; Note that different configuration select a different set of peaks, the user can handle FMA to take more or less peaks to optimize the feature extraction algorithms.

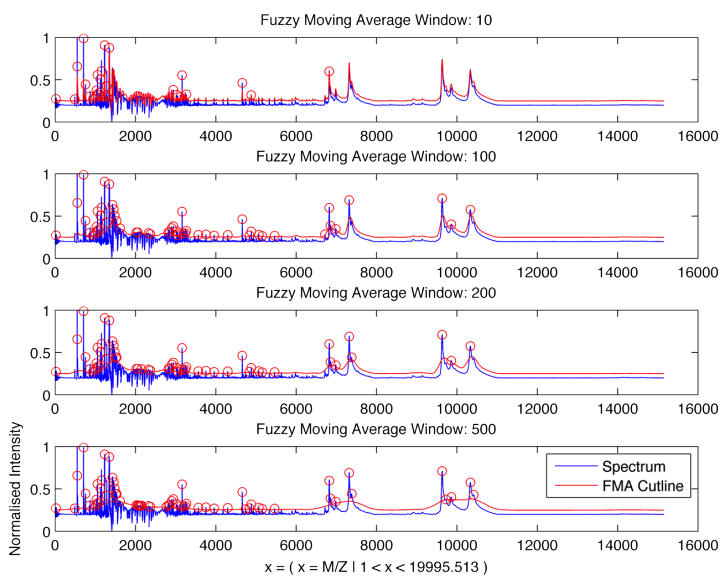



Figure 9: Representation of Fuzzy Moving average (FMA) with 10, 100, 200 and 500 moving window size. The blue line represent the spectra and the red line is the cut-line threshold.

*2.2. Feature Extraction*

The feature extraction is based on data transformation. To avoid overfitting on feature extraction process, on GA based feature extraction uses K-Fold Cross Validation as is showed on figure10.

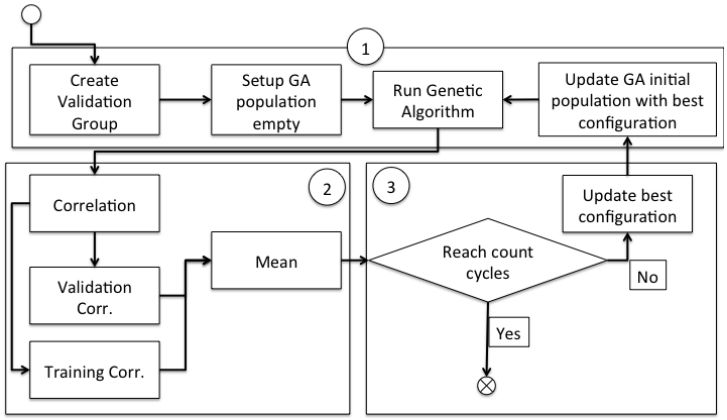



Figure 10: Abstract work-flow of all feature extraction cross validation functions. The figure section 1 represents the setup and execution of Genetic Algorithm as well creation of training and validation set. After the execution of Genetic Algorithm feature extractor, the figure section 2 represents the validation of feature extracted where "correlation step" measure the correlation value of selected features with its respectively classes for training and validation data sets. If the mean of training and validation correlations increase from past run the algorithm will assume that the current feature configuration is the best general feature. The figure section 3 represents the stop condition, that can be the cycles count or count of the same features generated consecutively.

*2.3. Best Peaks Selection*

The Best Peaks Selection is the procedure that chooses between the selected peaks the best ones. The figure 11 represent two methods of selection of best peaks, these methods is explained next :

- Correlation: Measure the Pearson correlation of position vector (Section: 2.1.2), on the peaks selection method, with its respectively classes generating the correlation spectrum where the 'x' axis represent the peaks positions and the y axis represents the correlation value with respective classes.

- Probability Based: Is based on Equation 7, that measure the peaks class exclusivity; that is a peak existence probability for each class.

$$D(x;i) = P(x_{(C_i,i)}) - \sum P(x_{(\not{C},i)}) \tag{6}$$

Having $C$ as the target class and $\not{C}$ (not C) the opposite; 'x' as peaks presence vector, one and zero represents the presence or absence of the peaks respectively; 'i' as 'x' index.

$$E(x) = \max(D(x)) \tag{7}$$

For both methods the user must choose an cut-line value; only peaks with values greater than cut-line is selected as best peak on algorithm selection process.

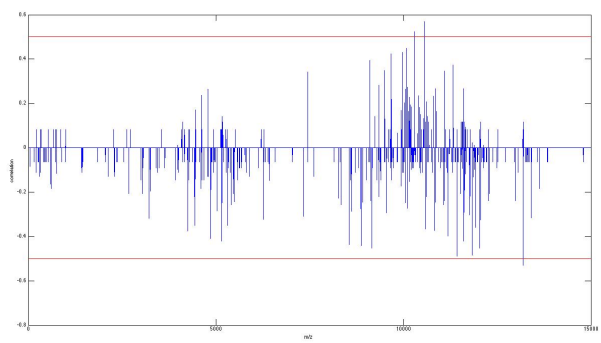



Figure 11: The correlation or probability spectrum is generated from the peaks presence on each spectra and its respective class. On probability spectrum is measured the probability of existence for all spectrum and calculate the difference for each class compared to the others.

### *2.4. Watch Points*

The watch points technique is focused to generate features to express the importance of neighborhood peaks. The watch points use fuzzy logic, where each watch point is a triangular weight window 12

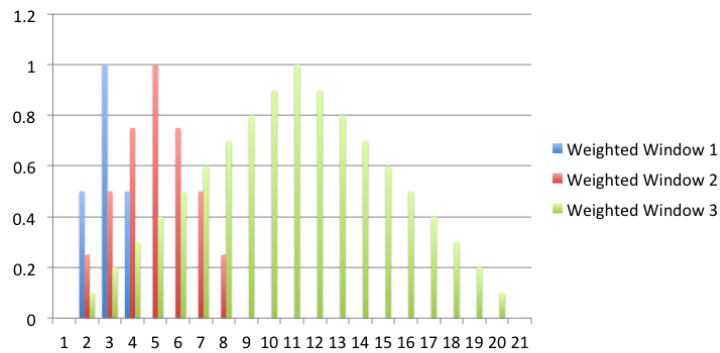



Figure 12: Representation of envelope of watch-points that is used on feature extraction process; those three watch-points have different centers and distances. Different distances imply on differences on weight granularity. The genetic algorithm changes the distances of all weighted windows seeking for correlated features.

On figure 12 considering three different watch points ( blue, red and green). An watch point have different size and weight distribution. The triangular function is expressed by following equation 8.

For L odd:

$$w(L,n) = \begin{cases} 2n/L+1 & 1 \leq n \leq (L+1)/2 \\ 2-2n/L+1 & (L+1)/2+1 \leq n \leq L \end{cases} \tag{8}$$

For L even:

$$w(L,n) = \begin{cases} (2n-1)/L & 1 \leq n \leq L/2 \\ 2-(2n-1)/L & L/2+1 \leq n \leq L \end{cases} \tag{9}$$

The watch point feature extraction process is based on the validation procedure showed on figure 10. The fitness function that leads the evolutionary algorithm is expressed on figure: 13

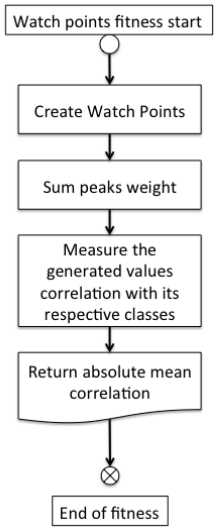



Figure 13: The Genetic algorithm watch point fitness workflow. this internal process generate a sum of all peaks weights relative to each watch-point and return an mean correlation of all watch points.

The best peaks selection process is used to choose the bests watch point centers but is the evolutionary algorithm GA which is responsible to determine the watch points distances. The feature extractor automatically optimizes the best distances for each selected centers. The second step on the fitness function is based on the sum of all weights. Considering that $w$ is the watch point universe expressed by the triangular function 8 on page 17 and $p$ is the position vector of peaks selection 2.1.2 the sum of peaks weights is leaded by following equation:

$$W = \sum w * p \tag{10}$$

Where $W$ is generated for each watch point and for spectra samples, where $w$ and $p$ has the same size and $w$ is a fuzzy triangle . The next step is the measure of all features correlation with its respectively classes and the result is the mean module correlation of all watch points.

*2.5. Feature Resonance*

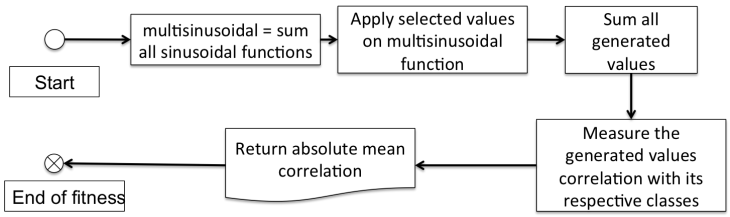



Figure 14: Representation of feature resonance fitness function. The multisinusoidal variable receive the sum of all sinusoidal functions; the graphic representation of multisinusoidal function is represented on figure 15 its general on equation 11 (Author 2015).

The Feature Resonance is also based on validation procedure. The main purpose of this feature extraction is to generate a wave with a sum of sinusoidal waves; this wave will be like a equalizer for each position on spectrum. The equalizer concept will be more clear visualizing the figure 15; it shows the sinusoidal function with one, two and three waves respectively and the red circles represents the peaks positions. The red circles may also be called as relevance value, that will be set for each point and summarized by equation 11.

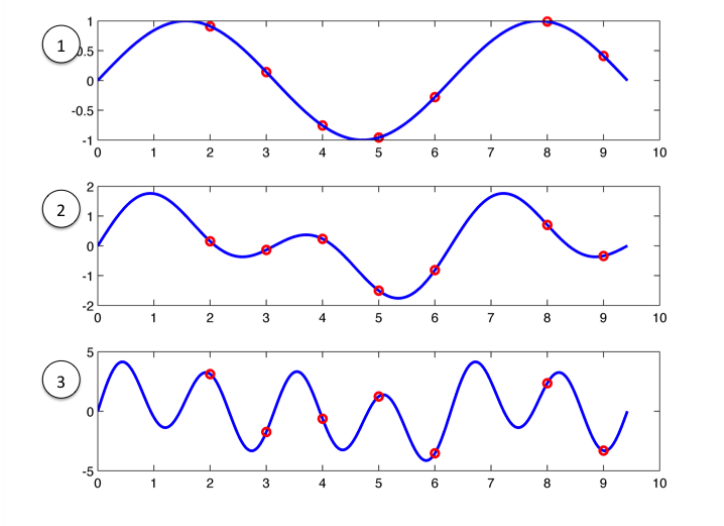



Figure 15: This is an simplified example of how feature resonance extractor works. The function 1 is a simple sinusoidal function where the red circles represent the function domain. The function 2 represent the sum of two sinusoidal functions and the function 3 represent an sum of three sinusoidal functions. Different configurations of frequency and amplitude can generate different values to the same peaks set.

The Feature Resonance can extract information of $m/z$ values or its respective intensity by the same way expressed on the following equation:

$$S(x; a, b, c) = \sum_{i=1}^{M} \sum (a_i * \sin(b_i + c_i * x)) \tag{11}$$

Where 'w' is the feature value; 'a', 'b' and 'c' are provided by the GA; 'x' is the provided va lues, 'M' is the number of waves set by user and $i\ E\ N^*$. Then the correlation value is measured and returned to the module of correlation, as is expressed on equation 12 to genetic algorithm .

$$|corr(S(x; a, b, c))| \tag{12}$$

There are some variants of the resonance feature extractor where the same algorithm could process the intensities and $m/z$ values. There is another variant that process only information over the specified the watch-points(Figure 16).

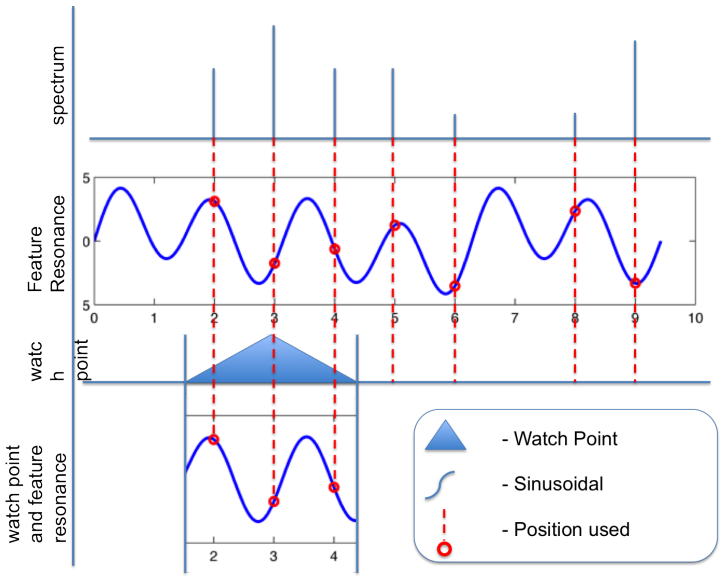



Figure 16: This is a summary of feature resonance working with the result of watch point feature extractor. Watch point and feature resonance is more performatic than the other ones but may acquire the same correlation performance; this combination can be used as a reinforcement feature for neural network.

While Feature Resonance feature extraction algorithm is running is possible to see the evolution of feature extraction; the figure 17 show the evolution weights generated by sinusoidal math function.

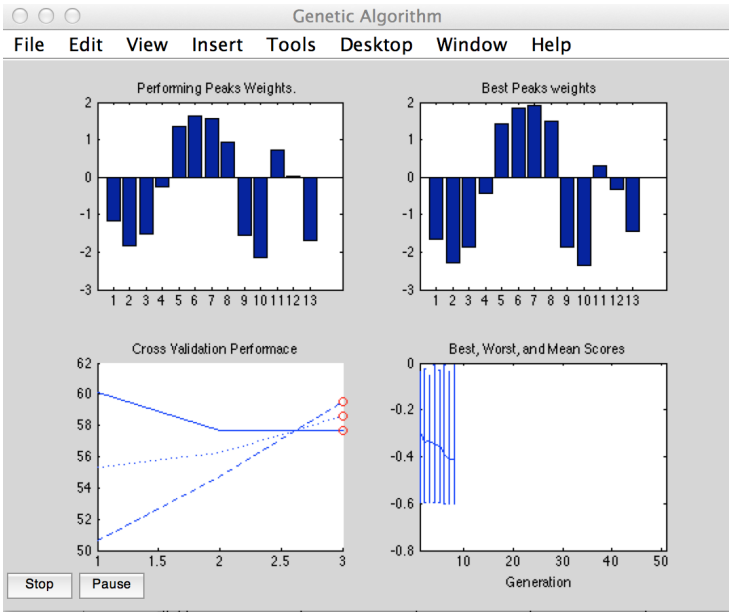



Figure 17: Show of Feature Resonance feature extraction process. The top left graph shows the current weights find by Genetic algorithm using current cross validation subset; the top right graph shows the best weights acquired on previous cycles. The bottom left graph show the track of correlation scores for training and validation subset and the bottom right graph show the range plot of all scores generated on this cycle.

## 3. Intact Cell Microorganism Results

The result of this research is MS-ML Studio, that is composed by an Matlab functions library, MS-ML Library, and an MS-ML User Interface represented on figure 18.

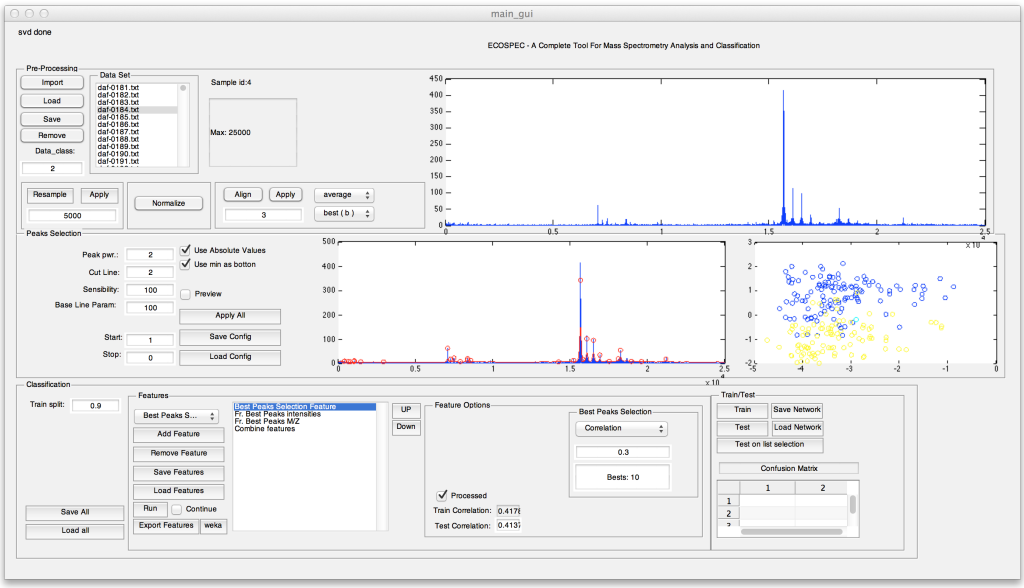



Figure 18: This figure shows the MS-ML User Interface. The source code of user interface is also provided as open source part of MS-ML Library. This interface is divided in four different main parts: Pre processing; Peaks selection; feature extraction and classification.

The MS-ML Studio (Figure 18) is open-source and is disponible on (http://goo.gl/l1x9iu). Using the MS-ML Studio is possible to handle all steps to extract features from spectra, train the neural network.

We done intact cell experiments with Microorganisms of Aeromonas family. All these experiments are made on Brucker MALDI-TOF analyzer. The data and bacterial sample preparation procedure and the original experiment can be found at Surek 2010 [19]. For identification experiment we use one against all strategy, were each class of microorganism is tested against the others. The samples used on these test contain four bacteria:

- A. veronii/sobria;
- A. cavie;
- A. hidrophilla
- A. trota

And the following groups was generated:

1. A. veronii/sobria against all;
2. A. cavie against all;
3. A. hidrophilla against all;
4. and A. trota against all

These four groups generate four experiments; for each experiment 90 percent of samples are used for the feature extraction and training task, the other 10% was isolated to test.

| | +R | −R | |
|---|---|---|---|
| | tp | fp | pp |
| | fn | tn | pn |
| | rp | rn | 1 |

| | +R | −R | |
|---|---|---|---|
| +P | A | B | A+B |
| −P | C | D | C+D |
| | A+C | B+D | N |

Figure 19: Systematic and traditional notations in a binary contingency table (Author: Powers 2011 [20]).

$$Accuracy = \frac{TP + TN}{N} \tag{13}$$

$$Precision = \frac{TP}{PP} \tag{14}$$

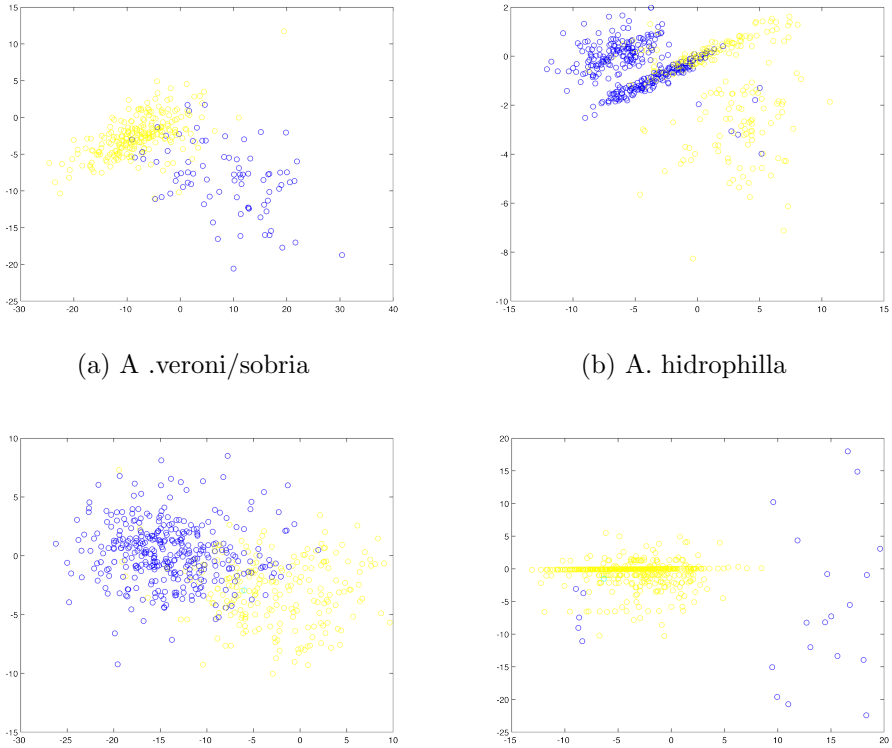
$$Recall = \frac{TP}{RP} \tag{15}$$

To measure the performance of training model we use Accuracy (Equation13) , Precision (Equation 14) and recall (Equation 15) equations as is defined by Powers 2011 [20] using as reference the Figure 19. The results for each group is represented on Table 3.

| Microorganisms | TR/TS | Recall | Precision | Accuracy |
|---|---|---|---|---|
| A. Hydrophilla | Training | 0.88856 | 0.95584 | 0.89824 |
| | Test | 0.89474 | 0.97143 | 0.91071 |
| A. Cavie | Training | 0.90169 | 0.83912 | 0.84344 |
| | Test | 0.83784 | 0.88571 | 0.82143 |
| A. Veronii/Sobria | Training | 0.95161 | 0.81944 | 0.94406 |
| | Test | 0.83333 | 0.71429 | 0.9 |
| A. Trota | Training | 1 | 0.7619 | 0.99022 |
| | Test | 1 | 1 | 1 |

Table 3: Precision measures achieved with ML-MS Studio for each four Aeromonas groups. Note that the data subset used for training was used exclusively for feature extraction and neural network training; the test subset was isolated to prove the performance on training set.

To prove that the feature extraction is efficient the blind test represent the test show the real performance of learning model. We create plots of two first principal component of PCA using ML-MS Studio and the extracted features with purpose to show graphically the performance of feature extraction. Is possible to see clearly that after feature extraction the information is much more representative.



(a) A .veroni/sobria

(b) A. hidrophilla

(c) A. cavie

(d) A. trota

Figure 20: PCA classification applied for all features extracted form each samples. The blue circles represent the target class, or class one.

### *3.1. A. veroni/sobria feature extraction configuration*

The group "A. veronii/sobria against all" we use: "FR Selected $M/Z$" resulting in features with 0.67 of correlation with classes, "FR Best Peaks Intensity" with correlation threshold of 0.18 generating feature values with 0.66 of correlation and "FR Best Peaks m/z" with correlation threshold of 0.12, 0.16, 0.2 and 0.25 generating features with correlation 0.77, 078, 0.48, 0.63. The Principal Component Analysis of two principal components of A.veroni/sobria feature extraction data-set is represented on Figure 20(a). Figures 21 and 21 shows the classification results of A. veroni/sobria spectra using embedded MLP neural network.

Confusion Matrix

| | 1 | 2 |
|---|---|---|
| 1 | 62 | 10 |
| 2 | 2 | 212 |

Figure 21: Confusion matrix for training data-set of A.veroni/sobria spectra feature extraction. Class one represent A.veroni/sobria and class 2 represent the other classes. Only ten of class two was classified as class one and only 2 was misclassified from class one. Total of spectra used on training task: 286.

Confusion Matrix

| | 1 | 2 |
|---|---|---|
| 1 | 4 | 3 |
| 2 | 0 | 23 |

Figure 22: Confusion matrix for test data-set of A.veroni/sobria spectra feature extraction. Class one represent A.veroni/sobria and class 2 represent the other classes. All A.veroni/sobria spectra was classified correctly and only three false positive. Total of spectra used for testing: 30.

### *3.2. A. hydrophilla feature extraction configuration*

For the "A. hydrophilla against all" group, we use FR. Best Peaks MZ Feature extraction with best peaks selection correlation threshold 0.11, 0.12, 0.15 and 0.16 generating features with correlation 0.66, 0.64, 0.57 and 0.56 for the training set. The Principal Component Analysis of two principal components of A.hydrophilla feature extraction data-set is represented on Figure 20(b) and the confusion matrix of training and test result is shown on Figures 23 and 24.

Confusion Matrix

| | 1 | 2 |
|---|---|---|
| 1 | 292 | 25 |
| 2 | 23 | 171 |

Figure 23: Confusion matrix for training data-set of A.hidrophilla spectra feature extraction. Class one represent A.hidrophilla and class 2 represent the other classes. twenty five spectra of class two was classified as class one and twenty three was misclassified from class one. Total of spectra used on training task: 511.

Confusion Matrix

| | 1 | 2 |
|---|---|---|
| 1 | 33 | 2 |
| 2 | 2 | 19 |

Figure 24: Confusion matrix for test data-set of A.hidropilla spectra feature extraction. Class one represent A.hidrophilla and class two represent the other classes. two class one spectra was misclassified and two spectra of class two was classified as one. Total of spectra used for testing: 56.

### *3.3. A. cavie feature extraction configuration*

For "A. cavie against all" group we use "FR. selected $M/Z$" feature extractor; "Fr. Best Peaks $M/Z$" with correlation threshold of 0.12, 0.16, 0.2 and "Fr. Best Peaks Intensities" with correlation threshold of 0.18. The Principal Component Analysis representation of two principal components for "A. cavie against all" is represented on Figure 20(c) and the confusion matrix is shown on Figures 25 and 26 .

Confusion Matrix

| | 1 | 2 |
|---|---|---|
| 1 | 296 | 21 |
| 2 | 29 | 165 |

Figure 25: Confusion matrix for training data-set of A.cavie spectra feature extraction. Class one represent A.cavie and class 2 represent the other classes. twenty five spectra of class two was classified as class one and twenty three was misclassified from class one. Total of spectra used on training task: 511.

Confusion Matrix

| | 1 | 2 |
|---|---|---|
| 1 | 33 | 2 |
| 2 | 4 | 17 |

Figure 26: Confusion matrix for test data-set of A.cavie spectra feature extraction. Class one represent A.cavie and class two represent the other classes. Two class one spectra was misclassified and four spectra of class two was classified as one. Total of spectra used for testing: 56.

### *3.4. A. trota feature extraction configuration*

On "A. trota against all" group we use "Fr. Selected $M/Z$" feature extraction generating feature with correlation 0.57 and "FR Best Peaks $M/Z$" with correlation threshold of 0.12, 0.2 and 0.4 generating features with correlation 0.33, 0.11 and 0.68. The Principal Component Analysis of two principal components of A.trota feature extraction data-set is represented on Figure 20(d) and on Figures 27 and 28 is shown the training and test results.

Confusion Matrix

| | 1 | 2 |
|---|---|---|
| 1 | 16 | 5 |
| 2 | 0 | 490 |

Figure 27: Confusion matrix for training data-set of A.trota spectra feature extraction. Class one represent A.trota and class 2 represent the other classes. twenty five spectra of class two was classified as class one and twenty three was misclassified from class one. Total of spectra used on training task: 511.

Confusion Matrix

| | 1 | 2 |
|---|---|---|
| 1 | 2 | 0 |
| 2 | 0 | 54 |

Figure 28: Confusion matrix for test data-set of A.trota spectra feature extraction. Class one represent A.trota and class two represent the other classes. No one of 56 spectra was misclassified.

## 4. Conclusion

The ML-MS Studio has been shown to analyze different mass spectrometry data from different approaches. Note that even with few spectra our feature extractor was capable acquire good results generating well correlated features. Future improvements on MS-ML will be developed to help the scientists visualize the peaks internal correlation to generate new analysis insights; new features extractors will be developed to improve performance of neural network. Future work we will apply MS-ML Studio with other identification problems.

## 5. References


[1] R. J. Arnold, N. Jayasankar, D. Aggarwal, H. Tang, P. Radivojac, A machine learning approach to predicting peptide fragmentation spectra., Pacific Symposium on Biocomputing. Pacific Symposium on Biocomputing (2006) 219–30.
URL http://www.ncbi.nlm.nih.gov/pubmed/17094241

[2] K. Bohme, I. C. Fernández-No, J. Barros-Velázquez, J. M. Gallardo, P. Carlo-Mata, B. Cañas, Species differentiation of seafood spoilage and pathogenic gram-negative bacteria by MALDI-TOF mass fingerprinting, Joural of proteome research (2010) 3169–3183.
URL http://pubs.acs.org/doi/abs/10.1021/pr100047q

[3] T. Krishnamurthy, P. L. Ross, Rapid identification of bacteria by direct matrix-assisted laser desorption/ionization mass spectrometric analysis of whole cells., Rapid communications in mass spectrometry : RCM 10 (15) (1996) 1992–6. doi:10.1002/(SICI)1097-0231(199612)10:15¡1992::AID-RCM789¿3.0.CO;2-V.
URL http://www.ncbi.nlm.nih.gov/pubmed/9004531

[4] A. Croxatto, G. Prod'hom, G. Greub, Applications of MALDI-TOF mass spectrometry in clinical diagnostic microbiology, FEMS Microbiology Reviews 36 (2012) 380–407. doi:10.1111/j.1574-6976.2011.00298.x.

[5] M. Wagner, D. Naik, A. Pothen, Protocols for disease classification from mass spectrometry data, Proteomics 3 (SEPTEMBER 2003) (2003) 1692–1698. doi:10.1002/pmic.200300519.

[6] T. P. Conrads, V. a. Fusaro, S. Ross, D. Johann, V. Rajapakse, B. a. Hitt, S. M. Steinberg, E. C. Kohn, D. a. Fishman, G. Whitely, J. C. Barrett, L. a. Liotta, E. F. Petricoin, T. D. Veenstra, High-resolution serum proteomic features for ovarian cancer detection., Endocrine-related cancer 11 (2) (2004) 163–78.
URL http://www.ncbi.nlm.nih.gov/pubmed/15163296

[7] B. W. Buchan, K. M. Riebe, N. a. Ledeboer, Comparison of the MALDI Biotyper system using Sepsityper specimen processing to routine microbiological methods for identification of bacteria from positive blood culture bottles., Journal of clinical microbiology 50 (2)

(2012) 346–52. doi:10.1128/JCM.05021-11.
URL http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=3264150&tool=pm

[8] C. Yang, Z. He, W. Yu, Comparison of public peak detection algorithms for MALDI mass spectrometry data analysis., BMC bioinformatics 10 (2009) 4. doi:10.1186/1471-2105-10-4.
URL http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=2631518&tool=pm

[9] T. Pluskal, S. Castillo, A. Villar-Briones, M. Oresic, MZmine 2: modular framework for processing, visualizing, and analyzing mass spectrometry-based molecular profile data., BMC bioinformatics 11 (2010) 395. doi:10.1186/1471-2105-11-395.

[10] G. Tomasi, F. Savorani, S. B. Engelsen, Icoshift: An effective tool for the alignment of chromatographic data, Journal of Chromatography A 1218 (43) (2011) 7832–7840. doi:10.1016/j.chroma.2011.08.086.
URL http://dx.doi.org/10.1016/j.chroma.2011.08.086

[11] R. Armananzas, Y. Saeys, I. Inza, M. Garcia-Torres, C. Bielza, Y. V. D. Peer, P. Larranaga, Peakbin Selection in Mass Spectrometry Data Using a Consensus Approach with Estimation of Distribution Algorithms, IEEE/ACM Transactions on Computational Biology and Bioinformatics 8 (3) (2011) 760–774. doi:10.1109/TCBB.2010.18.

[12] D. S. Wishart, C. Knox, A. C. Guo, R. Eisner, N. Young, B. Gautam, D. D. Hau, N. Psychogios, E. Dong, S. Bouatra, R. Mandal, I. Sinelnikov, J. Xia, L. Jia, J. a. Cruz, E. Lim, C. a. Sobsey, S. Shrivastava, P. Huang, P. Liu, L. Fang, J. Peng, R. Fradette, D. Cheng, D. Tzur, M. Clements, A. Lewis, A. De Souza, A. Zuniga, M. Dawe, Y. Xiong, D. Clive, R. Greiner, A. Nazyrova, R. Shaykhutdinov, L. Li, H. J. Vogel, I. Forsythei, HMDB: A knowledgebase for the human metabolome, Nucleic Acids Research 37 (October 2008) (2009) 603–610. doi:10.1093/nar/gkn810.

[13] H. Ogata, S. Goto, K. Sato, W. Fujibuchi, H. Bono, M. Kanehisa, KEGG: Kyoto encyclopedia of genes and genomes, Nucleic Acids Research 27 (1) (1999) 29–34. doi:10.1093/nar/27.1.29.

[14] C. a. Smith, G. O'Maille, E. J. Want, C. Qin, S. a. Trauger, T. R. Brandon, D. E. Custodio, R. Abagyan, G. Siuzdak, METLIN: a metabo-

lite mass spectral database., Therapeutic drug monitoring 27 (6) (2005) 747–751. doi:10.1097/01.ftd.0000179845.53213.39.

[15] Y. Wang, J. Xiao, T. O. Suzek, J. Zhang, J. Wang, S. H. Bryant, PubChem: A public information system for analyzing bioactivities of small molecules, Nucleic Acids Research 37 (June) (2009) 623–633. doi:10.1093/nar/gkp456.

[16] V. Law, C. Knox, Y. Djoumbou, T. Jewison, A. C. Guo, Y. Liu, A. MacIejewski, D. Arndt, M. Wilson, V. Neveu, A. Tang, G. Gabriel, C. Ly, S. Adamjee, Z. T. Dame, B. Han, Y. Zhou, D. S. Wishart, DrugBank 4.0: Shedding new light on drug metabolism, Nucleic Acids Research 42 (November 2013) (2014) 1091–1097. doi:10.1093/nar/gkt1068.

[17] Y. Liu, U. Aickelin, J. Feyereisl, L. G. Durrant, Wavelet feature extraction and genetic algorithm for biomarker detection in colorectal cancer data, Knowledge-Based Systems 37 (2013) 502–514. doi:10.1016/j.knosys.2012.09.011.
URL `http://dx.doi.org/10.1016/j.knosys.2012.09.011`

[18] J. S. Yu, S. Ongarello, R. Fiedler, X. W. Chen, G. Toffolo, C. Cobelli, Z. Trajanoski, Ovarian cancer identification based on dimensionality reduction for high-throughput mass spectrometry data, Bioinformatics 21 (10) (2005) 2200–2209. doi:10.1093/bioinformatics/bti370.

[19] M. Surek, B. S. Vizzotto, E. M. Souza, F. D. O. Pedrosa, C. B. Dallagassa, S. M. S. S. Farah, C. M. T. Fadel-Picheth, Identification and antimicrobial susceptibility of Aeromonas spp. isolated from stool samples of Brazilian subjects with diarrhoea and healthy controls, Journal of Medical Microbiology 59 (2010) 373–374. doi:10.1099/jmm.0.014258-0.

[20] D. M. W. Powers, EVALUATION: FROM PRECISION, RECALL AND F-MEASURE TO ROC, INFORMEDNESS, MARKEDNESS & CORRELATION, Journal of Machine Learning Technologies 2 (1) (2011) 37–63.